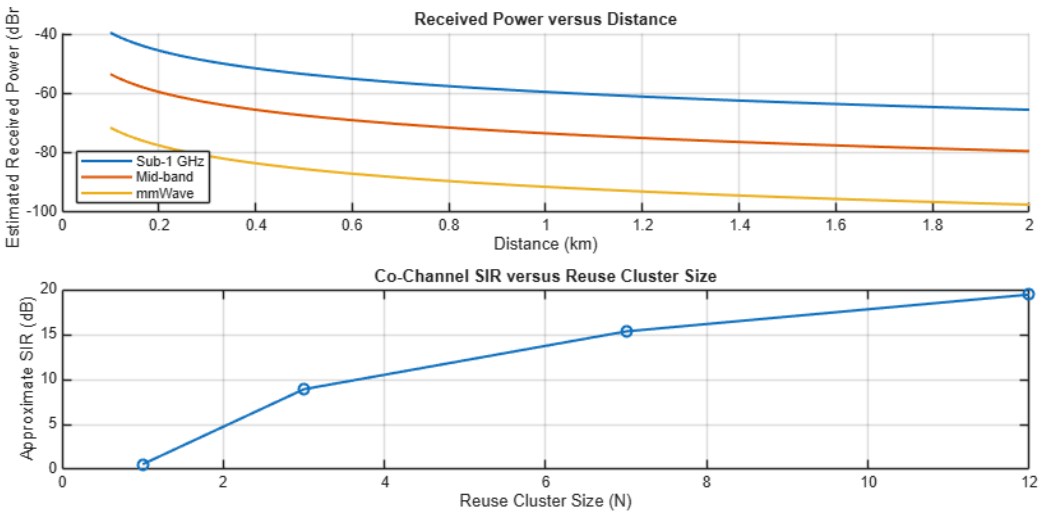
```
clc;
clear;
close all;

% Representative frequency bands
freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;

bandName = {'Sub-1 GHz'; 'Mid-band'; 'mmWave'};

% Link budget assumptions
txPowerdBm = 30;
gtDbi = 0;
grDbi = 0;

% Distance range in km
distanceKm = logspace(log10(0.1), log10(2), 200);
distanceAtPointKm = 0.5;

% Free-space path loss calculation
% Frequency in MHz and distance in km
fspl = @(f, d) 32.44 + 20*log10(f) + 20*log10(d);

% Calculate path loss at 0.5 km
lossAtPoint = zeros(length(freqMHz), 1);
rxAtPoint = zeros(length(freqMHz), 1);

for k = 1:length(freqMHz)
    lossAtPoint(k) = fspl(freqMHz(k), distanceAtPointKm);

    rxAtPoint(k) = txPowerdBm + gtDbi + grDbi ...
        - lossAtPoint(k);
end

% Display link budget results
linkTable = table(bandName, freqGHz(:), lossAtPoint, rxAtPoint, ...
    'VariableNames', ...
    {'Band', 'Frequency_GHz', 'FSPL_dB', 'EstimatedRx_dBm'});

disp('FREE-SPACE PATH LOSS AND RECEIVED POWER');
disp(linkTable);

% Plot 1: Free-space path loss
figure;
hold on;
grid on;

for k = 1:length(freqMHz)
    pathLoss = fspl(freqMHz(k), distanceKm);
    semilogx(distanceKm, pathLoss, 'LineWidth', 1.5);
end
```

```
xlabel('Distance (km)');
ylabel('Path Loss (dB)');
title('Free-Space Path Loss versus Distance');
legend(bandName, 'Location', 'northwest');
hold off;

% Plot 2: Estimated received power
figure;
hold on;
grid on;

for k = 1:length(freqMHz)
    rxPower = txPowerdBm + gtDbi + grDbi ...
        - fspl(freqMHz(k), distanceKm);

    semilogx(distanceKm, rxPower, 'LineWidth', 1.5);
end

xlabel('Distance (km)');
ylabel('Estimated Received Power (dBm)');
title('Received Power versus Distance');
legend(bandName, 'Location', 'southwest');
hold off;

% Co-channel frequency reuse calculation
cellRadiusKm = 0.5;
reuseN = [1 3 7 12];

pathLossExponent = 3.5;

% Approximate co-channel reuse distance
reuseDistance = sqrt(3 * reuseN) * cellRadiusKm;

% Approximate SIR assuming six equal-distance interferers
sirLinear = (reuseDistance ./ cellRadiusKm) ...
    .^ pathLossExponent / 6;

sirDb = 10 * log10(sirLinear);

% Display frequency reuse results
reuseTable = table(reuseN(:), reuseDistance(:), sirDb(:), ...
    'VariableNames', ...
    {'ReuseClusterN', 'ReuseDistance_km', 'ApproxSIR_dB'});

disp('CO-CHANNEL FREQUENCY REUSE RESULTS');
disp(reuseTable);

% Plot 3: SIR versus reuse cluster size
figure;
plot(reuseN, sirDb, 'o-', 'LineWidth', 1.5);
grid on;

xlabel('Reuse Cluster Size (N)');
ylabel('Approximate SIR (dB)');
```

```
title('Co-Channel SIR versus Reuse Cluster Size');

% Display received power at the selected distance
fprintf('\nReceived power at %.2f km:\n', distanceAtPointKm);

rxTable = table(bandName, rxAtPoint, ...
    'VariableNames', {'Band', 'ReceivedPower_dBm'});

disp(rxTable);

fprintf('\nAll calculations and graphs completed successfully.\n');
```

```
FREE-SPACE PATH LOSS AND RECEIVED POWER
        Band          Frequency_GHz    FSPL_dB    EstimatedRx_dBm
    ______________    _____________    _______    _______________

    {'Sub-1 GHz'}          0.7         83.321         -53.321
    {'Mid-band' }          3.5         97.301         -67.301
    {'mmWave'   }           28         115.36         -85.363

CO-CHANNEL FREQUENCY REUSE RESULTS
    ReuseClusterN    ReuseDistance_km    ApproxSIR_dB
    _____________    ________________    ____________

          1              0.86603          0.56811
          3                  1.5           8.9177
          7              2.2913            15.357
         12                   3            19.454


Received power at 0.50 km:
        Band          ReceivedPower_dBm
    ______________    _________________

    {'Sub-1 GHz'}          -53.321
    {'Mid-band' }          -67.301
    {'mmWave'   }          -85.363


All calculations and graphs completed successfully.
```



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